Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Rivercrest Cattle Co.

Physical address of dairy:

Number and Street	City	County	Zip Code
8600 Dos Rios LN	Modesto	Stanislaus	95358

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1990

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X017-X066-X006-XXXX

B. OPERATORS

Machado, Ed Jr.

Operator name: Machado, Ed Jr.

Telephone no.: (209) 599-2565 (Landline) (209) 481-0482 (Cellular)

Mailing Address Number and Street	City	State	Zip Code
11708 E Moncure RD	Ripon	CA	95366

Machado, Edward Anthony

Operator name: Machado, Edward Anthony

Telephone no.: (209) 599-2565 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
11708 E Moncure RD	Ripon	CA	95366

This operator is responsible for paying permit fees.

C. OWNERS

Machado, Edward Anthony

Legal owner name: Machado, Edward Anthony

Telephone no.: (209) 599-2565

Landline	Cellular
(209) 599-2565	

Mailing Address Number and Street	City	State	Zip Code
11708 E Moncure RD	Ripon	CA	95366

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	100	425	750	360	0
Number under roof	1,329	210	0	0	0	0
Maximum number	1,329	310	425	750	360	0
Average number	1,329	310	425	750	360	0
Avg live weight (lbs)	1,400	1,450	950	700		

Predominant milk cow breed: Holstein

Average milk production: 74 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 51,970.11 *tons per reporting period*

Total nitrogen from manure: 640,355.84 *lbs per reporting period*

After ammonia losses (30% loss applied): 448,249.09 *lbs per reporting period*

Total phosphorus from manure: 104,596.88 *lbs per reporting period*

Total potassium from manure: 258,008.37 *lbs per reporting period*

Total salt from manure: 697,044.15 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 33,198,000 *gallons*

Total nitrogen generated: 139,648.15 *lbs*

Total phosphorus generated: 21,437.65 *lbs*

Total potassium generated: 251,492.32 *lbs*

Total salt generated: 1,437,372.01 *lbs*

	33,198,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	33,198,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Ground water
San Joaquin River	Ground water
TID Lateral 2	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
03/02/2023	Solid commercial fertilizer *Can 17*	13.13 *ton*	Dry-weight	0.1	17.000000	0.000000	0.000000	0.000000
05/12/2023	Solid commercial fertilizer *4-0-0*	33.60 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000
05/14/2023	Solid commercial fertilizer *Ammonium Sulfate*	5.25 *ton*	Dry-weight	0.1	21.000000	0.000000	0.000000	0.000000
06/09/2023	Solid commercial fertilizer *UN 32*	18.10 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	20,920.26	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	20,920.26	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/15/2023	Corral solids	1,825.00 *ton*	Dry-weight	23.4		10,600.00	4,800.00	31,800.00		68.40
10/30/2023	Separator solids	6,225.00 *ton*	Dry-weight	77.4		14,600.00	1,930.00	6,700.00		6.60
11/02/2023	Corral solids	2,949.90 *ton*	Dry-weight	9.4		21,500.00	2,770.00	9,520.00		12.90

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	185,638.76	33,657.02	158,647.89	2,787,633.03
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	185,638.76	33,657.02	158,647.89	2,787,633.03

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	49	49	2	process wastewater	X017-X049-X002-XXXX
11	35	35	1	manure	X017-X060-X034-XXXX
12	9	9	2	manure	X017-X060-X034-XXXX
2	58	58	2	process wastewater	X017-X049-X002-XXXX
3	51	51	2	process wastewater	X017-X066-X006-XXXX
4	55	55	2	process wastewater	X017-X066-X006-XXXX
5	41	41	2	process wastewater	X017-X066-X006-XXXX
6	31	31	2	process wastewater	X017-X066-X006-XXXX
7	25	25	2	process wastewater	X017-X066-X006-XXXX
8	9	9	1	manure	X017-X066-X006-XXXX
9	4	4	1	manure	X017-X066-X006-XXXX
Field 13	42	40	1	manure	X017-X060-X004-XXXX
Field 14	17	17	2	manure	X017-X060-X004-XXXX
Totals for areas that were used for application	426	424	22		
Totals for areas that were not used for application					
Land application area totals	426	424	22		

B. CROPS AND HARVESTS

1

Field name: 1

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 49 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	852.60 *ton*	As-is		76.6	5,600.00	1,430.00	11,800.00		12.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	17.40	194.88	49.76	410.64	1,017.90

1

05/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 49 Plant date: 05/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/10/2023	1,342.60 *ton*	As-is		70.0	4,470.00	1,000.00	6,970.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.40	244.96	54.80	381.96	2,022.12

11

Field name: 11

02/07/2013: Almond Field Run

Crop: Almond Field Run Acres planted: 35 Plant date: 02/07/2013

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	242.40 *ton*	As-is		8.5	12,800.00	1,450.00	23,000.00		20.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	301.00	51.10	325.50	0.00
Total actual harvest content	6.93	177.30	20.08	318.58	2,598.18

12

Field name: 12

12

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 9 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	0.01 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 9 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	239.40 *ton*	As-is		70.0	4,420.00	910.00	5,440.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	26.60	235.14	48.41	289.41	1,611.96

2

Field name: 2

10/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 58 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	988.22 *ton*	As-is		74.8	5,970.00	1,520.00	13,200.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	17.04	203.44	51.80	449.81	1,013.30

2

05/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 58 Plant date: 05/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/10/2023	1,571.80 *ton*	As-is		70.0	4,610.00	770.00	5,980.00		13.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.10	249.86	41.73	324.12	2,195.10

3

Field name: 3

10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 51 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	856.80 *ton*	As-is		73.0	5,750.00	1,310.00	10,100.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	16.80	193.20	44.02	339.36	1,043.28

05/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 51 Plant date: 05/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	1,366.80 *ton*	As-is		70.0	4,830.00	760.00	5,770.00		12.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	26.80	258.89	40.74	309.27	2,010.00

4

Field name: 4

10/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 55 Plant date: 10/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	918.50 *ton*	As-is		75.4	5,660.00	1,360.00	11,400.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	16.70	189.04	45.42	380.76	994.18

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	1,501.50 *ton*	As-is		70.0	5,330.00	950.00	6,530.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.30	291.02	51.87	356.54	1,981.98

5

Field name: 5

11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 41 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	705.20 *ton*	As-is		75.4	5,610.00	1,380.00	11,900.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	17.20	192.98	47.47	409.36	1,032.41

5

05/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 41 Plant date: 05/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	1,082.40 *ton*	As-is		70.0	4,920.00	880.00	5,580.00		14.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	26.40	259.78	46.46	294.62	2,233.44

6

Field name: 6

11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 31 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	545.60 *ton*	As-is		77.8	4,550.00	1,190.00	9,740.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	17.60	160.16	41.89	342.85	890.84

05/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 31 Plant date: 05/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	877.20 *ton*	As-is		70.0	4,100.00	780.00	5,200.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.30	232.03	44.14	294.29	1,816.65

7

Field name: 7

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	437.50 *ton*	As-is		77.0	5,270.00	1,280.00	10,400.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	17.50	184.45	44.80	364.00	982.10

05/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 05/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	712.50 *ton*	As-is		70.0	4,670.00	920.00	6,190.00		9.68

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.50	266.19	52.44	352.83	1,655.28

8

Field name: 8

10/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 9 Plant date: 10/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/01/2023	0.01 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

9

Field name: 9

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 4 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	0.01 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

Field 13

Field name: Field 13

02/02/2019: Almond Field Run

Crop: Almond Field Run Acres planted: 40 Plant date: 02/02/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	256.10 *ton*	As-is		8.0	9,580.00	1,280.00	20,000.00		13.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	301.00	51.10	325.50	0.00
Total actual harvest content	6.40	122.67	16.39	256.10	1,590.38

Field 14

Field name: Field 14

Field 14

10/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 10/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	296.48 *ton*	As-is		77.6	4,770.00	1,300.00	11,000.00		12.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	17.44	166.38	45.34	383.68	1,000.08

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	459.30 *ton*	As-is		70.0	4,420.00	910.00	5,440.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.02	238.84	49.17	293.95	1,637.27

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 10/20/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.22	7.12	109.76	958.23	511,500.00 *gal*
Canal	Ground water	0.00	0.00	0.00	10.30	6,045,000.00 *gal*
Application event totals		51.22	7.12	109.76	968.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	133.20	22.09	230.99	907.08	990,000.00 *gal*
Application event totals		133.20	22.09	230.99	907.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.35	12.08	58.73	407.46	825,000.00 *gal*
Application event totals		30.35	12.08	58.73	407.46	

1 - 05/12/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 05/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	20.92	6,825,000.00 *gal*
Application event totals		0.00	0.00	0.00	20.92	

1 - 05/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	20.32	6,630,000.00 *gal*
Application event totals		0.00	0.00	0.00	20.32	

06/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.78	5.04	63.30	354.29	528,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	19.13	6,240,000.00 *gal*
Application event totals		36.78	5.04	63.30	373.42	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.78	5.04	63.30	354.29	528,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	19.13	6,240,000.00 *gal*
Application event totals		36.78	5.04	63.30	373.42	

1 - 05/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.78	5.04	63.30	354.29	528,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	19.13	6,240,000.00 *gal*
Application event totals		36.78	5.04	63.30	373.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.78	5.04	63.30	354.29	528,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	19.13	6,240,000.00 *gal*
Application event totals		36.78	5.04	63.30	373.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.78	5.04	63.30	354.29	528,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.53	6,045,000.00 *gal*
Application event totals		36.78	5.04	63.30	372.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.48	4.72	59.35	332.15	495,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	21.00	6,850,000.00 *gal*
Application event totals		34.48	4.72	59.35	353.15	

11 - 02/07/2013: Almond Field Run

Field name: 11

Crop: Almond Field Run

Plant date: 02/07/2013

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

11 - 02/07/2013: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	9.33	3,913,000.00 *gal*
Application event totals		0.00	0.00	0.00	9.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	9.33	3,913,000.00 *gal*
Application event totals		0.00	0.00	0.00	9.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	89.88	38.41	230.49	1,206.79	70.00 *ton*
Application event totals		89.88	38.41	230.49	1,206.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	29.75	0.00	0.00	0.00	
Application event totals		29.75	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	29.75	0.00	0.00	0.00	
TID Lateral 2	Ground water	0.00	0.00	0.00	15.21	3,543,750.00 *gal*
Application event totals		29.75	0.00	0.00	15.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.21	3,543,750.00 *gal*
Application event totals		0.00	0.00	0.00	15.21	

11 - 02/07/2013: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Amonium Sulfate	Solid commercial fertilizer	29.40	0.00	0.00	0.00	
TID Lateral 2	Ground water	0.00	0.00	0.00	15.21	3,543,750.00 *gal*
Application event totals		29.40	0.00	0.00	15.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.21	3,543,750.00 *gal*
Application event totals		0.00	0.00	0.00	15.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.37	3,580,500.00 *gal*
Application event totals		0.00	0.00	0.00	15.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.37	3,580,500.00 *gal*
Application event totals		0.00	0.00	0.00	15.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.37	3,580,500.00 *gal*
Application event totals		0.00	0.00	0.00	15.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.37	3,580,500.00 *gal*
Application event totals		0.00	0.00	0.00	15.37	

11 - 02/07/2013: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.37	3,580,500.00 *gal*
Application event totals		0.00	0.00	0.00	15.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.37	3,580,500.00 *gal*
Application event totals		0.00	0.00	0.00	15.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.37	3,580,500.00 *gal*
Application event totals		0.00	0.00	0.00	15.37	

12 - 11/02/2022: Oats, silage-soft dough

Field name: 12

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Corral solids	199.42	34.21	74.36	986.96	130.00 *ton*
Application event totals		199.42	34.21	74.36	986.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	10.85	1,170,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.85	

12 - 06/11/2023: Corn, silage

Field name: 12

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	21.15	1,267,500.00 *gal*
Application event totals		0.00	0.00	0.00	21.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	216.52	98.05	649.57	13,971.84	120.00 *ton*
Application event totals		216.52	98.05	649.57	13,971.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	19.53	1,170,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	19.53	1,170,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.53	

12 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	19.53	1,170,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	19.53	1,170,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	17.90	1,072,500.00 *gal*
Application event totals		0.00	0.00	0.00	17.90	

2 - 10/21/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.25	6.98	107.69	940.11	594,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	10.10	7,020,000.00 *gal*
Application event totals		50.25	6.98	107.69	950.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.19	9.89	152.55	1,331.82	841,500.00 *gal*
Application event totals		71.19	9.89	152.55	1,331.82	

2 - 10/21/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.03	17.42	182.13	715.24	924,000.00 *gal*
Application event totals		105.03	17.42	182.13	715.24	

2 - 05/13/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 05/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	21.21	8,190,000.00 *gal*
Application event totals		0.00	0.00	0.00	21.21	

05/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/10/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	19.70	7,605,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.70	

2 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.90	5.05	63.51	355.44	627,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	19.19	7,410,000.00 *gal*
Application event totals		36.90	5.05	63.51	374.63	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.90	5.05	63.51	355.44	627,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	19.19	7,410,000.00 *gal*
Application event totals		36.90	5.05	63.51	374.63	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.90	5.05	63.51	355.44	627,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	19.19	7,410,000.00 *gal*
Application event totals		36.90	5.05	63.51	374.63	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.90	5.05	63.51	355.44	627,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	19.19	7,410,000.00 *gal*
Application event totals		36.90	5.05	63.51	374.63	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.93	4.92	61.84	346.08	610,500.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.69	7,215,000.00 *gal*
Application event totals		35.93	4.92	61.84	364.77	

2 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.95	4.79	60.17	336.73	594,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.18	7,020,000.00 *gal*
Application event totals		34.95	4.79	60.17	354.91	

3 - 10/22/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.39	7.28	112.26	980.05	544,500.00 *gal*
Canal	Ground water	0.00	0.00	0.00	10.53	6,435,000.00 *gal*
Application event totals		52.39	7.28	112.26	990.58	

12/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	82.55	11.47	176.89	1,544.32	858,000.00 *gal*
Application event totals		82.55	11.47	176.89	1,544.32	

03/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	102.38	16.98	177.54	697.21	792,000.00 *gal*
Application event totals		102.38	16.98	177.54	697.21	

3 - 05/14/2023: Corn, silage

3 - 05/14/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 05/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.24	7.66	37.24	258.38	544,500.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.95	6,435,000.00 *gal*
Application event totals		19.24	7.66	37.24	277.33	

05/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	19.53	6,630,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.53	

06/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.34	4.84	60.82	340.40	528,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.38	6,240,000.00 *gal*
Application event totals		35.34	4.84	60.82	358.78	

3 - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.34	4.84	60.82	340.40	528,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.38	6,240,000.00 *gal*
Application event totals		35.34	4.84	60.82	358.78	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	19.53	6,630,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.53	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.34	4.84	60.82	340.40	528,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.38	6,240,000.00 *gal*
Application event totals		35.34	4.84	60.82	358.78	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.23	4.69	58.92	329.76	511,500.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	17.80	6,045,000.00 *gal*
Application event totals		34.23	4.69	58.92	347.56	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	17.80	6,045,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.80	

3 - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.23	4.69	58.92	329.76	511,500.00 *gal*
Application event totals		34.23	4.69	58.92	329.76	

4 - 10/23/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough

Plant date: 10/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	10.36	6,825,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.36	

10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.52	7.16	110.40	963.85	577,500.00 *gal*
Application event totals		51.52	7.16	110.40	963.85	

12/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.66	9.82	151.41	1,321.85	792,000.00 *gal*
Application event totals		70.66	9.82	151.41	1,321.85	

03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	106.80	17.71	185.21	727.32	891,000.00 *gal*
Application event totals		106.80	17.71	185.21	727.32	

4 - 05/15/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.47	7.75	37.67	261.37	594,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	19.17	7,020,000.00 *gal*
Application event totals		19.47	7.75	37.67	280.54	

05/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	20.24	7,410,000.00 *gal*
Application event totals		0.00	0.00	0.00	20.24	

06/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.84	4.91	61.69	345.23	577,500.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.64	6,825,000.00 *gal*
Application event totals		35.84	4.91	61.69	363.87	

4 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.84	4.91	61.69	345.23	577,500.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.64	6,825,000.00 *gal*
Application event totals		35.84	4.91	61.69	363.87	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	19.17	7,020,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.17	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.84	4.91	61.69	345.23	577,500.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.64	6,825,000.00 *gal*
Application event totals		35.84	4.91	61.69	363.87	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.81	4.77	59.92	335.37	561,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.11	6,630,000.00 *gal*
Application event totals		34.81	4.77	59.92	353.48	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.81	4.77	59.92	335.37	561,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.11	6,630,000.00 *gal*
Application event totals		34.81	4.77	59.92	353.48	

5 - 11/06/2022: Oats, silage-soft dough

5 - 11/06/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough

Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	116.74	19.36	202.44	794.99	726,000.00 *gal*
Application event totals		116.74	19.36	202.44	794.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	122.04	20.24	211.64	831.13	759,000.00 *gal*
Application event totals		122.04	20.24	211.64	831.13	

5 - 05/11/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 05/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	21.43	5,850,000.00 *gal*
Application event totals		0.00	0.00	0.00	21.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

5 - 05/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	20.00	5,460,000.00 *gal*
Application event totals		0.00	0.00	0.00	20.00	

06/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.71	4.89	61.47	344.03	429,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.57	5,070,000.00 *gal*
Application event totals		35.71	4.89	61.47	362.60	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.92	10.53	132.40	740.99	924,000.00 *gal*
San Joaquin River	Ground water	2.32	0.00	0.00	418.88	4,200,000.00 *gal*
Application event totals		79.24	10.53	132.40	1,159.87	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.71	4.89	61.47	344.03	429,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.57	5,070,000.00 *gal*
Application event totals		35.71	4.89	61.47	362.60	

5 - 05/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.71	4.89	61.47	344.03	429,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.57	5,070,000.00 *gal*
Application event totals		35.71	4.89	61.47	362.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.34	4.70	59.11	330.80	412,500.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	17.86	4,875,000.00 *gal*
Application event totals		34.34	4.70	59.11	348.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.97	4.51	56.74	317.57	396,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	17.15	4,680,000.00 *gal*
Application event totals		32.97	4.51	56.74	334.71	

6 - 11/06/2022: Oats, silage-soft dough

Field name: 6

Crop: Oats, silage-soft dough

Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	119.30	19.78	206.89	812.47	561,000.00 *gal*
Application event totals		119.30	19.78	206.89	812.47	

6 - 11/06/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.27	17.46	182.55	716.89	495,000.00 *gal*
Application event totals		105.27	17.46	182.55	716.89	

6 - 05/11/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 05/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	21.73	4,485,000.00 *gal*
Application event totals		0.00	0.00	0.00	21.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	19.84	4,095,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

6 - 05/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.33	4.97	62.54	350.01	330,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.90	3,900,000.00 *gal*
Application event totals		36.33	4.97	62.54	368.90	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.12	10.70	134.46	752.51	709,500.00 *gal*
San Joaquin River	Ground water	2.35	0.00	0.00	425.39	3,225,000.00 *gal*
Application event totals		80.47	10.70	134.46	1,177.90	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.52	4.73	59.41	332.51	313,500.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	17.95	3,705,000.00 *gal*
Application event totals		34.52	4.73	59.41	350.46	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.52	4.73	59.41	332.51	313,500.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	17.95	3,705,000.00 *gal*
Application event totals		34.52	4.73	59.41	350.46	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.52	4.73	59.41	332.51	313,500.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	17.95	3,705,000.00 *gal*
Application event totals		34.52	4.73	59.41	350.46	

6 - 05/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.70	4.48	56.29	315.00	297,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	17.01	3,510,000.00 *gal*
Application event totals		32.70	4.48	56.29	332.01	

7 - 11/07/2022: Oats, silage-soft dough

Field name: 7

Crop: Oats, silage-soft dough Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	121.83	20.20	211.28	829.68	462,000.00 *gal*
Application event totals		121.83	20.20	211.28	829.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	121.83	20.20	211.28	829.68	462,000.00 *gal*
Application event totals		121.83	20.20	211.28	829.68	

7 - 05/11/2023: Corn, silage

Field name: 7

Crop: Corn, silage Plant date: 05/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

7 - 05/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	21.09	3,510,000.00 *gal*
Application event totals		0.00	0.00	0.00	21.09	

05/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	19.92	3,315,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.92	

06/08/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.04	4.93	62.04	347.21	264,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.75	3,120,000.00 *gal*
Application event totals		36.04	4.93	62.04	365.95	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.59	10.49	131.83	737.81	561,000.00 *gal*
San Joaquin River	Ground water	2.31	0.00	0.00	417.08	2,550,000.00 *gal*
Application event totals		78.90	10.49	131.83	1,154.89	

7 - 05/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.04	4.93	62.04	347.21	264,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.75	3,120,000.00 *gal*
Application event totals		36.04	4.93	62.04	365.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.04	4.93	62.04	347.21	264,000.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	18.75	3,120,000.00 *gal*
Application event totals		36.04	4.93	62.04	365.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.79	4.63	58.16	325.51	247,500.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	17.57	2,925,000.00 *gal*
Application event totals		33.79	4.63	58.16	343.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.79	4.63	58.16	325.51	247,500.00 *gal*
TID Lateral 2	Ground water	0.00	0.00	0.00	17.57	2,925,000.00 *gal*
Application event totals		33.79	4.63	58.16	343.08	

8 - 10/06/2022: Oats, silage-soft dough

Field name: 8

Crop: Oats, silage-soft dough

Plant date: 10/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

8 - 10/06/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Corral solids	199.42	34.21	74.36	986.96	130.00 *ton*
Application event totals		199.42	34.21	74.36	986.96	

10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	10.76	0.00	0.00	767.74	1,200,000.00 *gal*
Application event totals		10.76	0.00	0.00	767.74	

9 - 10/27/2022: Oats, silage-soft dough

Field name: 9

Crop: Oats, silage-soft dough Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Corral solids	207.09	35.52	77.22	1,024.92	60.00 *ton*
Application event totals		207.09	35.52	77.22	1,024.92	

10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
San Joaquin River	Ground water	10.59	0.00	0.00	755.74	525,000.00 *gal*
Application event totals		10.59	0.00	0.00	755.74	

Field 13 - 02/02/2019: Almond Field Run

Field name: Field 13

Crop: Almond Field Run Plant date: 02/02/2019

Field 13 - 02/02/2019: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	9.28	4,448,000.00 *gal*
Application event totals		0.00	0.00	0.00	9.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	9.28	4,448,000.00 *gal*
Application event totals		0.00	0.00	0.00	9.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	67.41	28.81	172.86	905.09	60.00 *ton*
Application event totals		67.41	28.81	172.86	905.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	29.75	0.00	0.00	0.00	
Application event totals		29.75	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	8.37	4,010,000.00 *gal*
Application event totals		0.00	0.00	0.00	8.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	29.75	0.00	0.00	0.00	
Application event totals		29.75	0.00	0.00	0.00	

Field 13 - 02/02/2019: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.06	4,010,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.06	4,010,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Amonium Sulfate	Solid commercial fertilizer	29.40	0.00	0.00	0.00	
Application event totals		29.40	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.06	4,010,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.25	4,060,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.25	4,060,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.25	

Field 13 - 02/02/2019: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.25	4,060,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.25	4,060,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.25	4,060,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.25	4,060,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	15.25	4,060,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.25	

Field 14 - 10/23/2022: Oats, silage-soft dough

Field name: Field 14

Crop: Oats, silage-soft dough

Plant date: 10/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 14 - 10/23/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Corral solids	194.91	33.43	72.68	964.63	240.00 *ton*
Application event totals		194.91	33.43	72.68	964.63	

10/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	11.01	2,242,500.00 *gal*
Application event totals		0.00	0.00	0.00	11.01	

Field 14 - 06/11/2023: Corn, silage

Field name: Field 14

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	20.68	2,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	20.68	

05/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	219.71	99.49	65.91	14,177.31	230.00 *ton*
Application event totals		219.71	99.49	65.91	14,177.31	

06/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Field 14 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	18.95	2,145,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.95	

07/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	18.95	2,145,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.95	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	18.95	2,145,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.95	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	18.95	2,145,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.95	

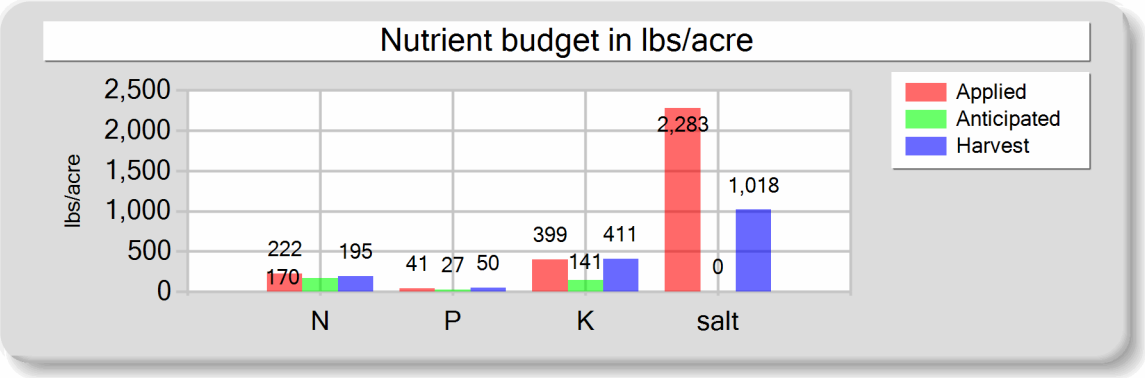
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Ground water	0.00	0.00	0.00	18.09	2,047,500.00 *gal*
Application event totals		0.00	0.00	0.00	18.09	

B. NUTRIENT BUDGET

1 - 10/20/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	214.77	41.29	399.48	2,272.77
Fresh water	0.00	0.00	0.00	10.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	221.77	41.29	399.48	2,283.06
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	194.88	49.76	410.64	1,017.90
Nutrient balance	26.89	-8.48	-11.16	1,265.16
Applied to removed ratio	1.14	0.83	0.97	2.24

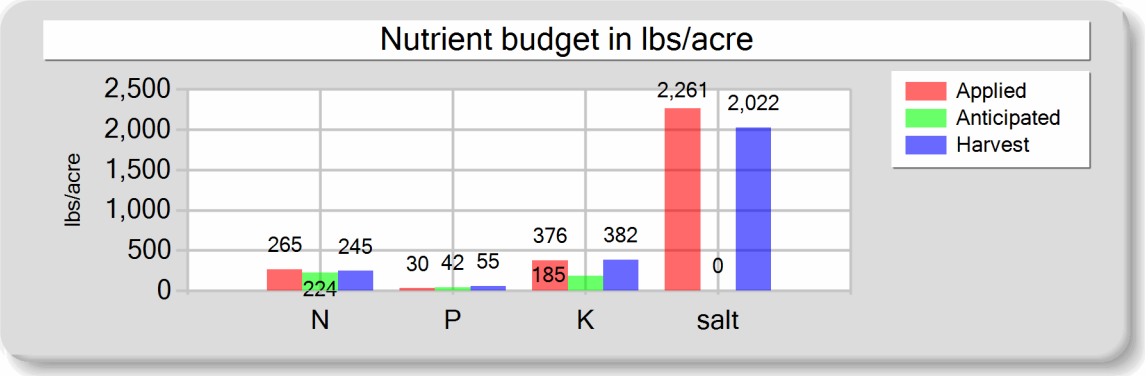
Fresh water applied	
6,045,000.00	*gallons*
222.62	*acre-inches*
4.54	*inches/acre*

Process wastewater applied	
2,326,500.00	*gallons*
85.68	*acre-inches*
1.75	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 05/12/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 05/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	218.37	29.90	375.87	2,103.60
Fresh water	0.00	0.00	0.00	157.29
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	265.37	29.90	375.87	2,260.90
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	244.96	54.80	381.96	2,022.12
Nutrient balance	20.41	-24.90	-6.08	238.78
Applied to removed ratio	1.08	0.55	0.98	1.12

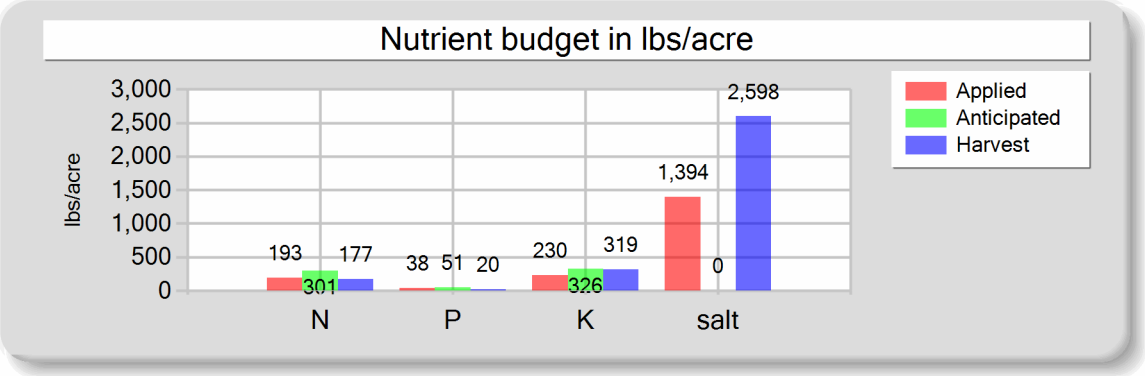
Fresh water applied	
51,310,000.00	*gallons*
1,889.57	*acre-inches*
38.56	*inches/acre*

Process wastewater applied	
3,135,000.00	*gallons*
115.45	*acre-inches*
2.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 02/07/2013: Almond Field Run

Field name: 11 Crop: Almond Field Run Plant date: 02/07/2013



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	88.90	0.00	0.00	0.00
Dry manure	89.88	38.41	230.49	1,206.79
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	187.06
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	192.78	38.41	230.49	1,393.85
Anticipated crop nutrient removal	301.00	51.10	325.50	0.00
Actual crop nutrient removal	177.30	20.08	318.58	2,598.18
Nutrient balance	15.48	18.33	-88.10	-1,204.33
Applied to removed ratio	1.09	1.91	0.72	0.54

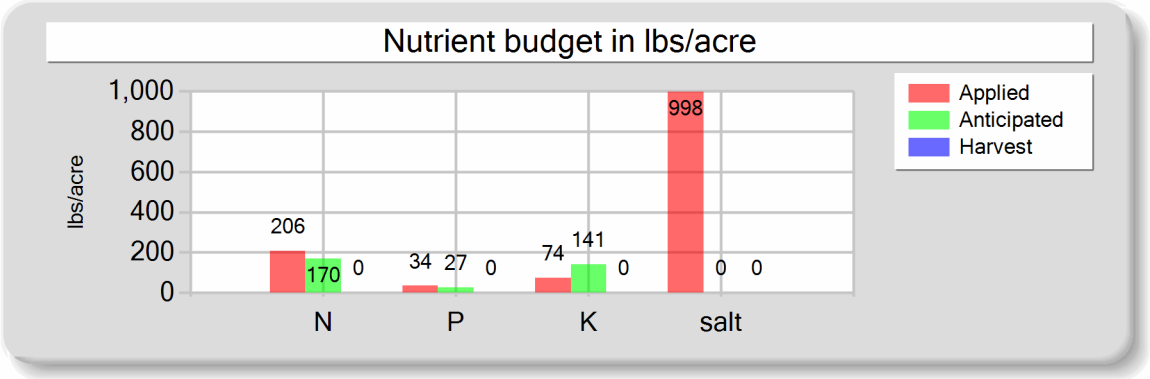
Fresh water applied	
47,064,500.00	*gallons*
1,733.23	*acre-inches*
49.52	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

12 - 11/02/2022: Oats, silage-soft dough

Field name: 12 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	199.42	34.21	74.36	986.96
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	10.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.42	34.21	74.36	997.81
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	206.42	34.21	74.36	997.81
Applied to removed ratio	0.00	0.00	0.00	0.00

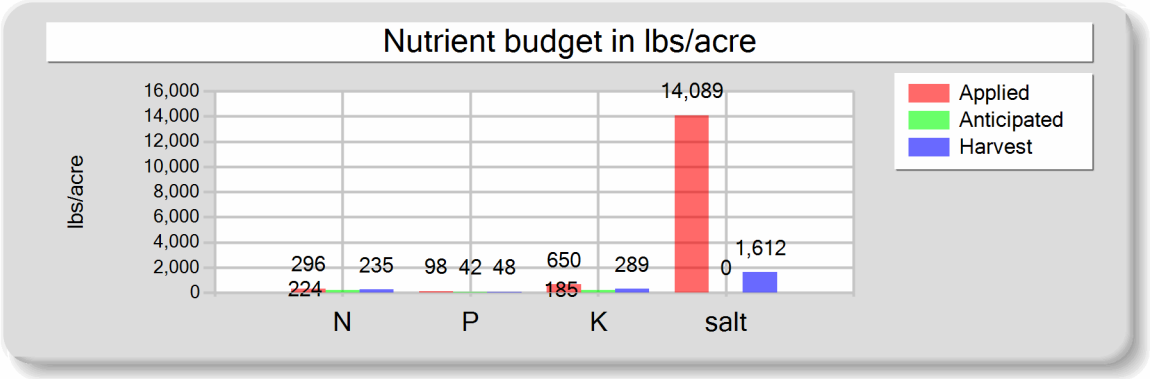
Fresh water applied	
1,170,000.00	*gallons*
43.09	*acre-inches*
4.79	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

12 - 06/11/2023: Corn, silage

Field name: 12 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	72.00	0.00	0.00	0.00
Dry manure	216.52	98.05	649.57	13,971.84
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	117.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	295.52	98.05	649.57	14,089.00
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	235.14	48.41	289.41	1,611.96
Nutrient balance	60.38	49.64	360.16	12,477.04
Applied to removed ratio	1.26	2.03	2.24	8.74

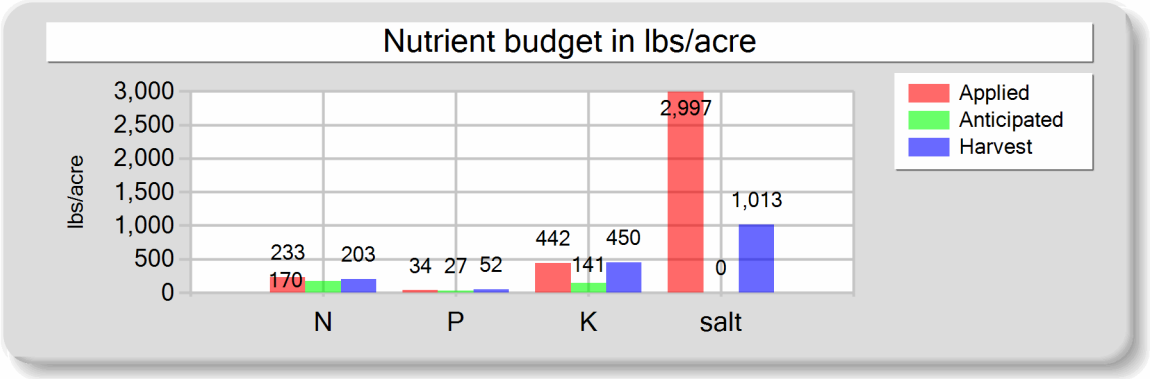
Fresh water applied	
7,020,000.00	*gallons*
258.52	*acre-inches*
28.72	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 10/21/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	226.47	34.29	442.37	2,987.17
Fresh water	0.00	0.00	0.00	10.10
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	233.47	34.29	442.37	2,997.27
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	203.44	51.80	449.81	1,013.30
Nutrient balance	30.03	-17.51	-7.44	1,983.97
Applied to removed ratio	1.15	0.66	0.98	2.96

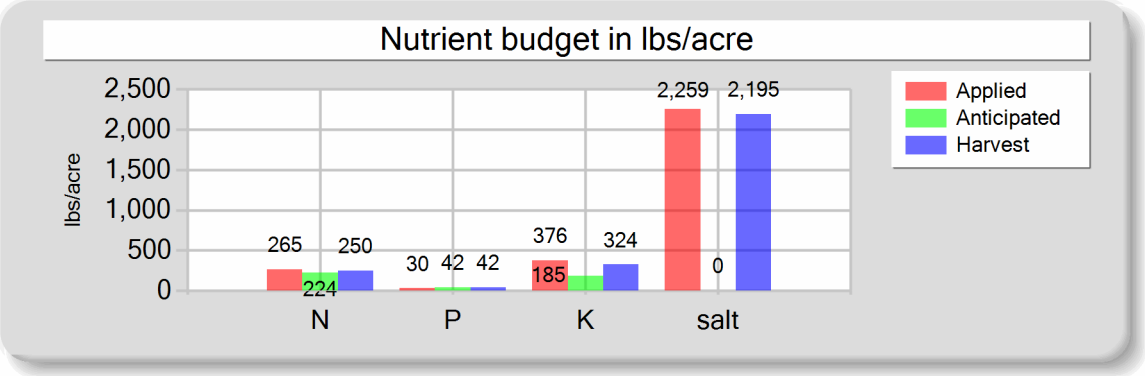
Fresh water applied	
7,020,000.00	*gallons*
258.52	*acre-inches*
4.46	*inches/acre*

Process wastewater applied	
2,359,500.00	*gallons*
86.89	*acre-inches*
1.50	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 05/13/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 05/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	218.47	29.91	376.04	2,104.56
Fresh water	0.00	0.00	0.00	154.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	265.47	29.91	376.04	2,259.09
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	249.86	41.73	324.12	2,195.10
Nutrient balance	15.61	-11.82	51.93	63.99
Applied to removed ratio	1.06	0.72	1.16	1.03

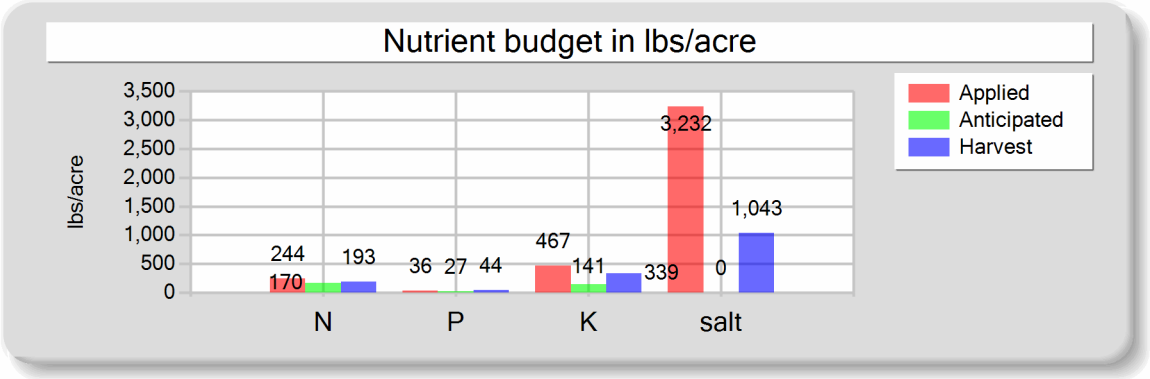
Fresh water applied	
59,670,000.00	*gallons*
2,197.44	*acre-inches*
37.89	*inches/acre*

Process wastewater applied	
3,712,500.00	*gallons*
136.72	*acre-inches*
2.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 10/22/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	237.32	35.73	466.70	3,221.57
Fresh water	0.00	0.00	0.00	10.53
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	244.32	35.73	466.70	3,232.10
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	193.20	44.02	339.36	1,043.28
Nutrient balance	51.12	-8.29	127.34	2,188.82
Applied to removed ratio	1.26	0.81	1.38	3.10

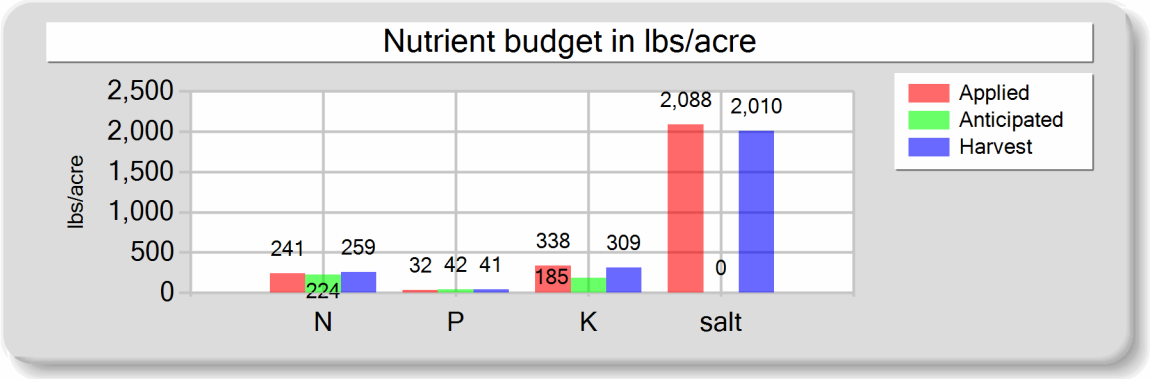
Fresh water applied	
6,435,000.00	*gallons*
236.98	*acre-inches*
4.65	*inches/acre*

Process wastewater applied	
2,194,500.00	*gallons*
80.82	*acre-inches*
1.58	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 05/14/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 05/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	193.71	31.55	337.55	1,939.09
Fresh water	0.00	0.00	0.00	148.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.71	31.55	337.55	2,087.84
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	258.89	40.74	309.27	2,010.00
Nutrient balance	-18.17	-9.19	28.28	77.84
Applied to removed ratio	0.93	0.77	1.09	1.04

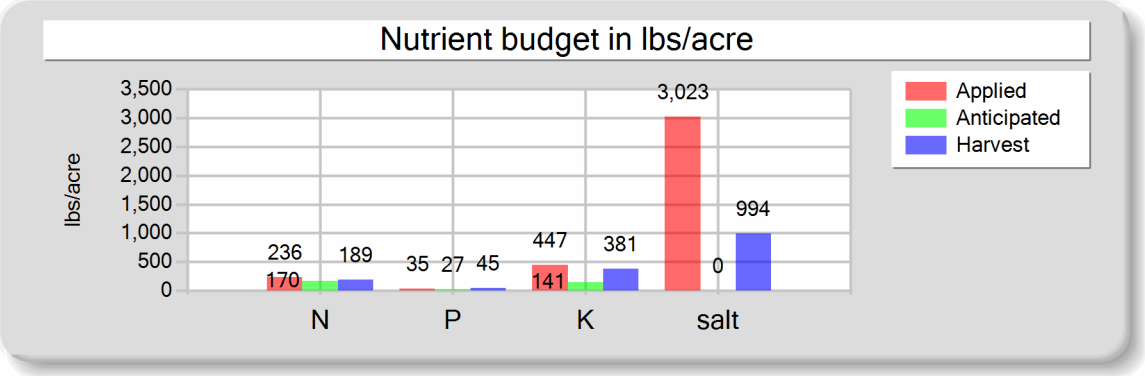
Fresh water applied	
50,505,000.00	*gallons*
1,859.93	*acre-inches*
36.47	*inches/acre*

Process wastewater applied	
3,151,500.00	*gallons*
116.06	*acre-inches*
2.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 10/23/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 10/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.98	34.69	447.02	3,013.01
Fresh water	0.00	0.00	0.00	10.36
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.98	34.69	447.02	3,023.37
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	189.04	45.42	380.76	994.18
Nutrient balance	46.94	-10.74	66.26	2,029.18
Applied to removed ratio	1.25	0.76	1.17	3.04

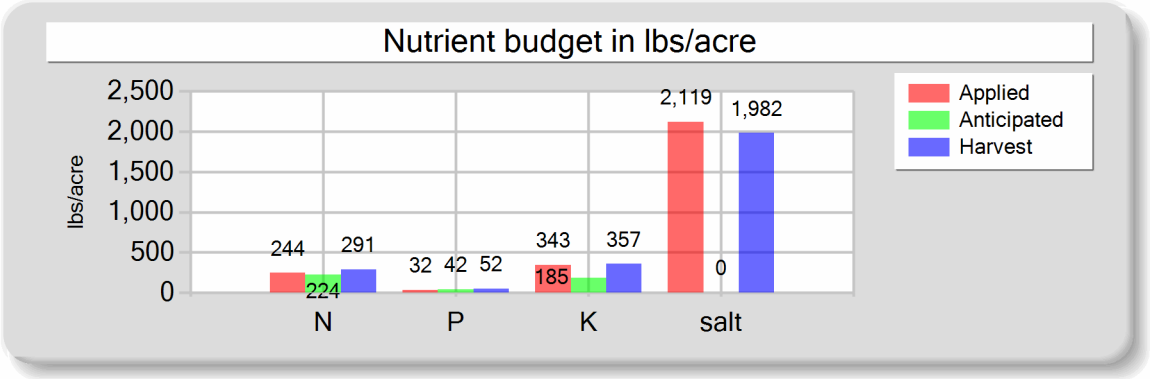
Fresh water applied	
6,825,000.00	*gallons*
251.34	*acre-inches*
4.57	*inches/acre*

Process wastewater applied	
2,260,500.00	*gallons*
83.25	*acre-inches*
1.51	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 05/15/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	196.61	32.00	342.58	1,967.80
Fresh water	0.00	0.00	0.00	150.72
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	243.61	32.00	342.58	2,118.52
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	291.02	51.87	356.54	1,981.98
Nutrient balance	-47.41	-19.87	-13.96	136.54
Applied to removed ratio	0.84	0.62	0.96	1.07

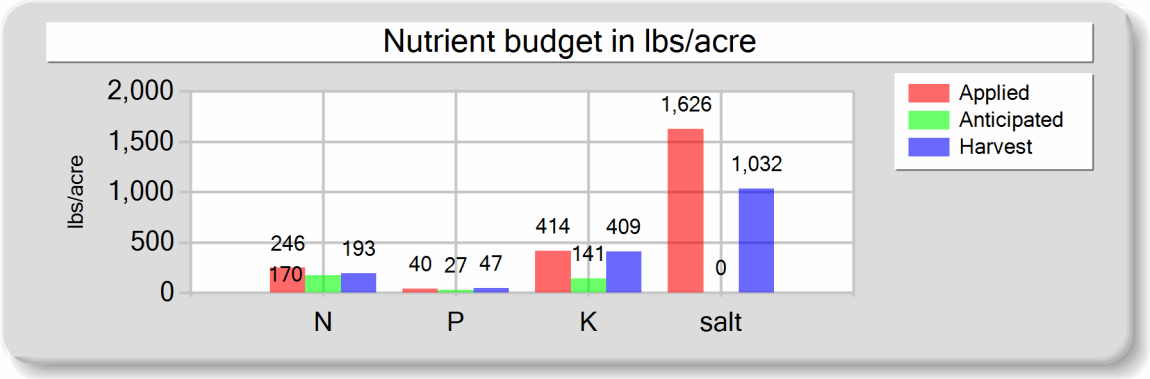
Fresh water applied	
55,185,000.00	*gallons*
2,032.28	*acre-inches*
36.95	*inches/acre*

Process wastewater applied	
3,448,500.00	*gallons*
127.00	*acre-inches*
2.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 11/06/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	238.78	39.59	414.09	1,626.11
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.78	39.59	414.09	1,626.11
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	192.98	47.47	409.36	1,032.41
Nutrient balance	52.79	-7.88	4.73	593.70
Applied to removed ratio	1.27	0.83	1.01	1.58

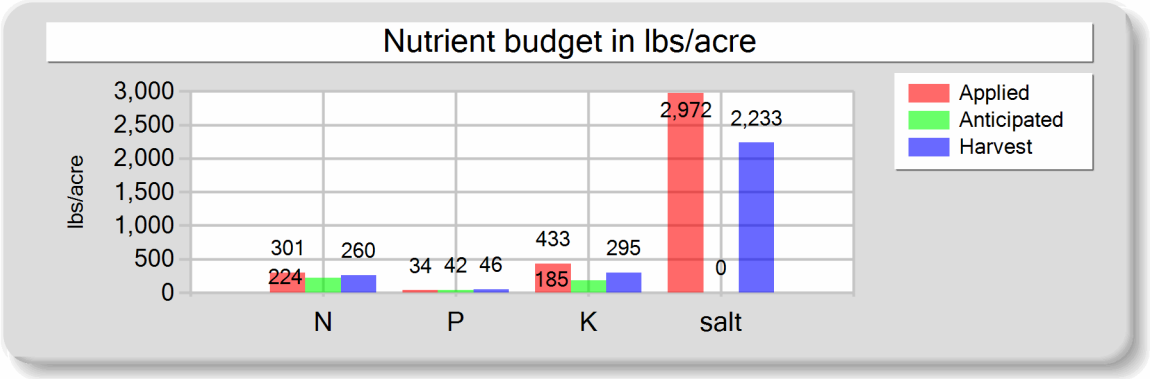
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,485,000.00 *gallons*
54.69 *acre-inches*
1.33 *inches/acre*

Total harvests for the crop
1 *harvests*

5 - 05/11/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 05/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	251.36	34.42	432.66	2,421.44
Fresh water	2.32	0.00	0.00	551.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	300.68	34.42	432.66	2,972.48
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	259.78	46.46	294.62	2,233.44
Nutrient balance	40.90	-12.05	138.04	739.04
Applied to removed ratio	1.16	0.74	1.47	1.33

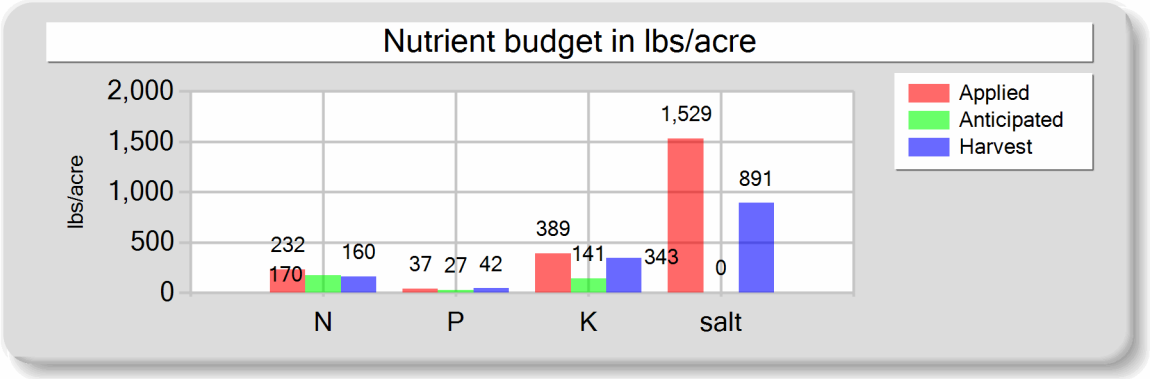
Fresh water applied	
40,275,000.00	*gallons*
1,483.19	*acre-inches*
36.18	*inches/acre*

Process wastewater applied	
3,019,500.00	*gallons*
111.20	*acre-inches*
2.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 11/06/2022: Oats, silage-soft dough

Field name: 6 Crop: Oats, silage-soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	224.57	37.24	389.45	1,529.36
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	231.57	37.24	389.45	1,529.36
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	160.16	41.89	342.85	890.84
Nutrient balance	71.41	-4.65	46.60	638.52
Applied to removed ratio	1.45	0.89	1.14	1.72

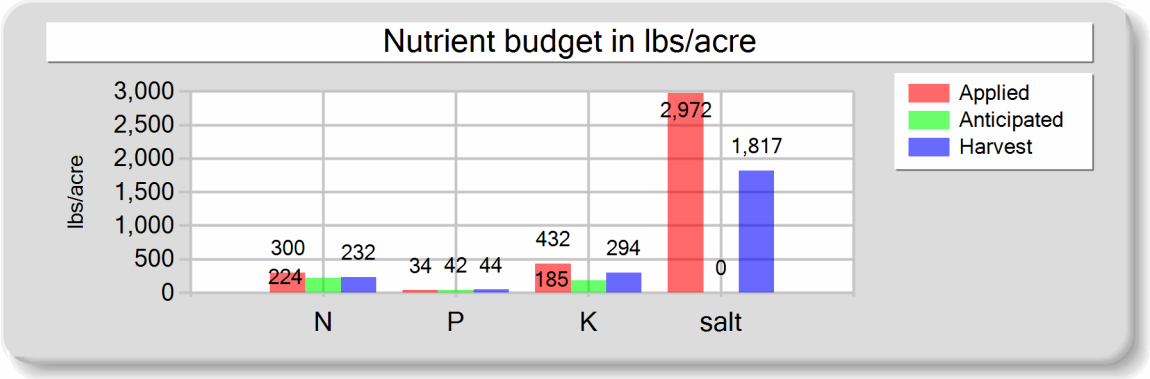
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,056,000.00 *gallons*
38.89 *acre-inches*
1.25 *inches/acre*

Total harvests for the crop
1 *harvests*

6 - 05/11/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 05/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	250.70	34.33	431.52	2,415.04
Fresh water	2.35	0.00	0.00	556.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	300.05	34.33	431.52	2,971.77
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	232.03	44.14	294.29	1,816.65
Nutrient balance	68.02	-9.82	137.23	1,155.11
Applied to removed ratio	1.29	0.78	1.47	1.64

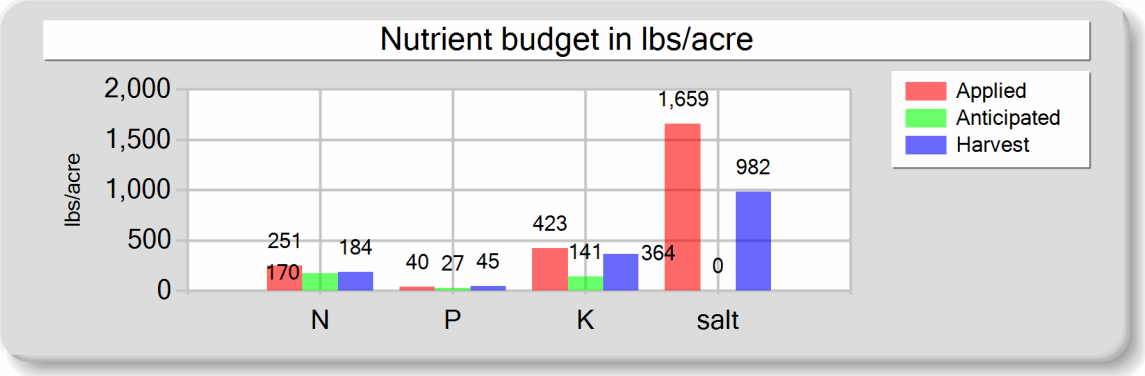
Fresh water applied	
30,330,000.00	*gallons*
1,116.95	*acre-inches*
36.03	*inches/acre*

Process wastewater applied	
2,277,000.00	*gallons*
83.85	*acre-inches*
2.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 11/07/2022: Oats, silage-soft dough

Field name: 7 Crop: Oats, silage-soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	243.66	40.40	422.55	1,659.36
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	250.66	40.40	422.55	1,659.36
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	184.45	44.80	364.00	982.10
Nutrient balance	66.21	-4.40	58.55	677.26
Applied to removed ratio	1.36	0.90	1.16	1.69

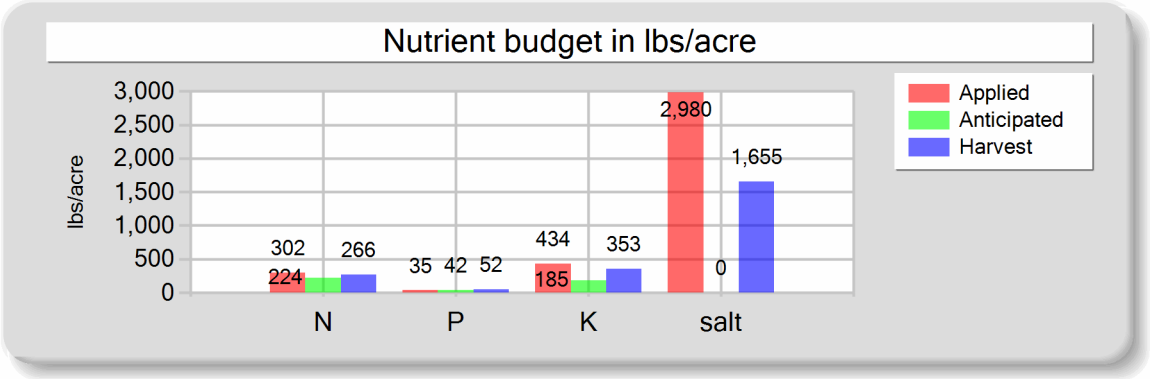
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
924,000.00	*gallons*
34.03	*acre-inches*
1.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 05/11/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 05/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	252.30	34.54	434.27	2,430.44
Fresh water	2.31	0.00	0.00	549.48
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	301.60	34.54	434.27	2,979.92
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	266.19	52.44	352.83	1,655.28
Nutrient balance	35.41	-17.90	81.44	1,324.64
Applied to removed ratio	1.13	0.66	1.23	1.80

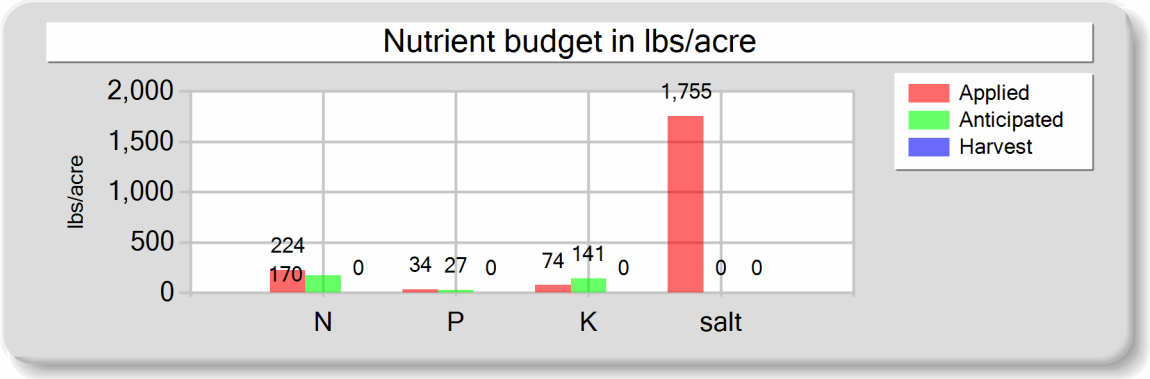
Fresh water applied	
24,585,000.00	*gallons*
905.38	*acre-inches*
36.22	*inches/acre*

Process wastewater applied	
1,848,000.00	*gallons*
68.06	*acre-inches*
2.72	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 10/06/2022: Oats, silage-soft dough

Field name: 8 Crop: Oats, silage-soft dough Plant date: 10/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	199.42	34.21	74.36	986.96
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	10.76	0.00	0.00	767.74
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	224.18	34.21	74.36	1,754.70
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	224.18	34.21	74.36	1,754.70
Applied to removed ratio	0.00	0.00	0.00	0.00

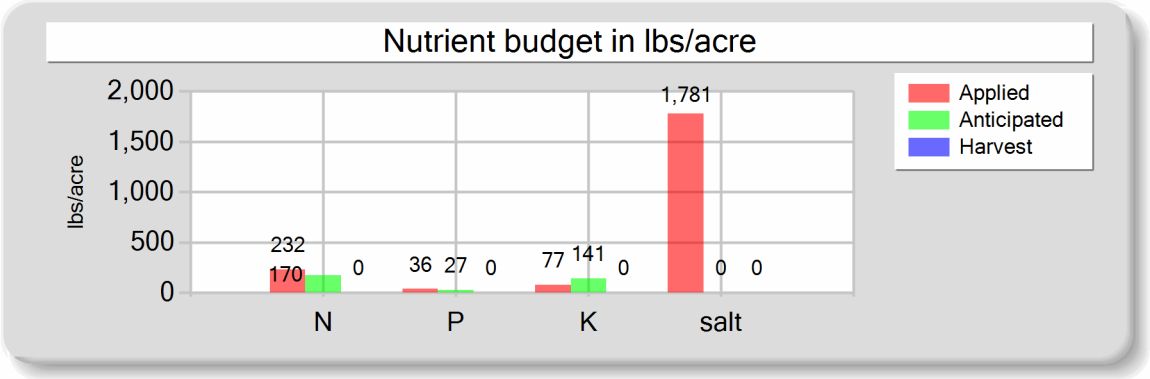
Fresh water applied	
1,200,000.00	*gallons*
44.19	*acre-inches*
4.91	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 10/27/2022: Oats, silage-soft dough

Field name: 9 Crop: Oats, silage-soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	207.09	35.52	77.22	1,024.92
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	10.59	0.00	0.00	755.74
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	231.68	35.52	77.22	1,780.66
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	231.68	35.52	77.22	1,780.66
Applied to removed ratio	0.00	0.00	0.00	0.00

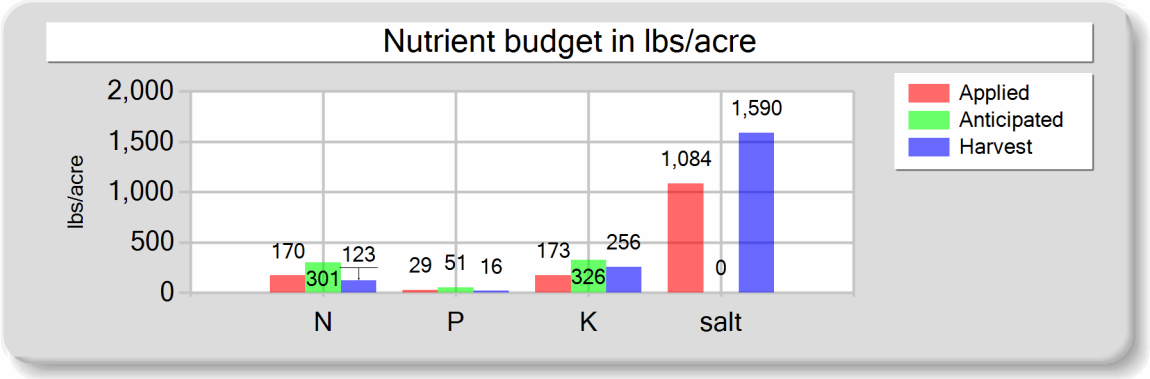
Fresh water applied	
525,000.00	*gallons*
19.33	*acre-inches*
4.83	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 13 - 02/02/2019: Almond Field Run

Field name: Field 13 Crop: Almond Field Run Plant date: 02/02/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	88.90	0.00	0.00	0.00
Dry manure	67.41	28.81	172.86	905.09
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	178.83
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	170.31	28.81	172.86	1,083.92
Anticipated crop nutrient removal	301.00	51.10	325.50	0.00
Actual crop nutrient removal	122.67	16.39	256.10	1,590.38
Nutrient balance	47.63	12.42	-83.24	-506.46
Applied to removed ratio	1.39	1.76	0.67	0.68

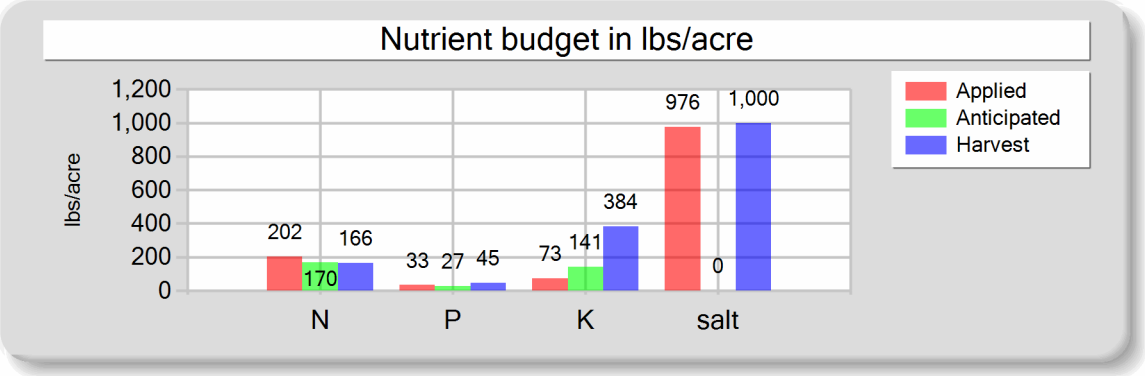
Fresh water applied	
53,356,000.00	*gallons*
1,964.92	*acre-inches*
49.12	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 14 - 10/23/2022: Oats, silage-soft dough

Field name: Field 14 Crop: Oats, silage-soft dough Plant date: 10/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	194.91	33.43	72.68	964.63
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	11.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.91	33.43	72.68	975.64
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	166.38	45.34	383.68	1,000.08
Nutrient balance	35.53	-11.91	-311.00	-24.44
Applied to removed ratio	1.21	0.74	0.19	0.98

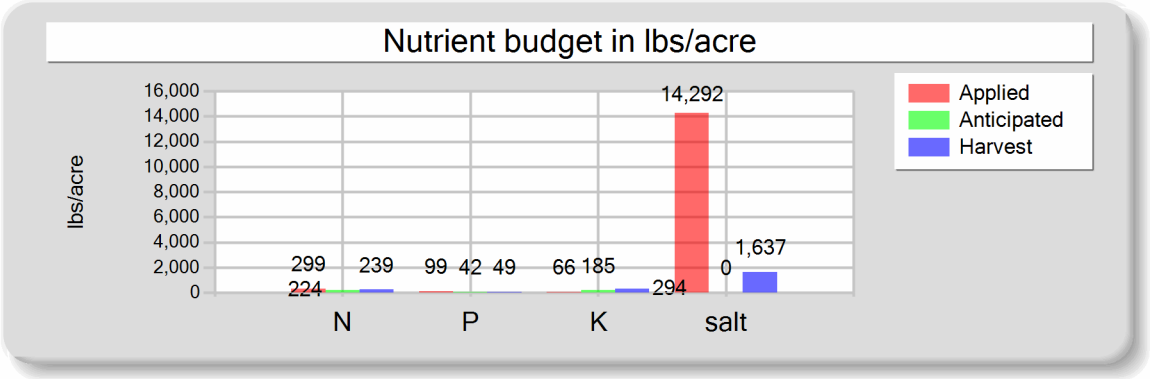
Fresh water applied	
2,242,500.00	*gallons*
82.58	*acre-inches*
4.86	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 14 - 06/11/2023: Corn, silage

Field name: Field 14 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	72.00	0.00	0.00	0.00
Dry manure	219.71	99.49	65.91	14,177.31
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	114.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	298.71	99.49	65.91	14,291.89
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	238.84	49.17	293.95	1,637.27
Nutrient balance	59.87	50.32	-228.04	12,654.62
Applied to removed ratio	1.25	2.02	0.22	8.73

Fresh water applied	
12,967,500.00	*gallons*
477.55	*acre-inches*
28.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 9.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,800.00	10,600.00	63,600.00							33.30
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 09/14/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,500.00	5,060.00	11,000.00							14.60
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/21/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 23.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,600.00	4,800.00	31,800.00							68.40
DL	100.00	100.00	100.00							0.01

Separated Solids

Sample and source description: Separated Solids

Sample date: 04/21/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,600.00	3,900.00	13,900.00							6.63
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 9.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,500.00	2,770.00	9,520.00	13,200.00	4,040.00	2,400.00	7,640.00	890.00		12.90
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 09/12/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	1,930.00	6,700.00							6.60
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 10/20/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	588.00	311.00	0.00	0.00	81.70	1,260.00								9,410.00	11,000
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	790.00	417.00	0.00	0.00	131.00	1,370.00								8,950.00	5,380
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	790.00	417.00	0.00	0.00	131.00	1,370.00								8,950.00	5,380
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 04/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	216.00	126.00	0.00	0.00	86.00	418.00								4,260.00	2,900
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 04/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	216.00	126.00	0.00	0.00	86.00	418.00								4,260.00	2,900
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	409.00	202.00	0.00	0.00	56.00	704.00								5,700.00	3,940
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 11/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	739.00	451.00	0.00	0.00	96.80	1,630.00								9,900.00	7,920
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

Canal

Fresh Water

Sample description: Fresh Water

Sample date: 10/05/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								100.00	10
DL	0.50	0.50	0.50								1.00	10

San Joaquin River

Fresh Water

Sample description: Fresh Water

Sample date: 11/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	27,100.00	0.00	27,100.00								1,040.00	490
DL	0.50	0.50	0.50								1.00	10

TID Lateral 2

Fresh Water

Sample description: Fresh Water

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.00	18
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

1

Soils

Sample and source description: Soils

Sample date: 11/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			84.00				
DL			4.00				

11

Soils

Sample and source description: Soils

Sample date: 11/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			20.00				
DL			4.00				

12

Soils

Sample and source description: Soils

Sample date: 11/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			20.00				
DL			4.00				

2

2

Soils

Sample and source description: Soils

Sample date: 11/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			66.00				
DL			4.00				

3

Soils

Sample and source description: Soils

Sample date: 11/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			86.00				
DL			4.00				

4

Soils

Sample and source description: Soils

Sample date: 11/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			75.00				
DL			4.00				

5

5

Soils

Sample and source description: Soils

Sample date: 11/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			34.00				
DL			4.00				

6

Soils

Sample and source description: Soils

Sample date: 11/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			51.00				
DL			4.00				

7

Soils

Sample and source description: Soils

Sample date: 11/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			42.00				
DL			4.00				

8

8

Soils

Sample and source description: Soils

Sample date: 11/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			86.00				
DL			4.00				

9

Soils

Sample and source description: Soils

Sample date: 11/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			86.00				
DL			4.00				

Field 13

Soils

Sample and source description: Soils

Sample date: 11/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			15.00				
DL			4.00				

Field 14

Field 14

Soils

Sample and source description: Soils

Sample date: 11/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			15.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

1 - 10/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	1,430.00	11,800.00		12.50
DL	100.00	100.00	100.00		0.01

1 - 05/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,470.00	1,000.00	6,970.00		12.30
DL	100.00	100.00	100.00		0.01

1 - 05/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,370.00	950.00	7,020.00		11.70
DL	100.00	100.00	100.00		0.01

11 - 02/07/2013: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	1,450.00	23,000.00		20.50
DL	100.00	100.00	100.00		0.01

2 - 10/21/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,970.00	1,520.00	13,200.00		11.80
DL	100.00	100.00	100.00		0.01

2 - 05/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,160.00	850.00	5,560.00		10.60
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,610.00	770.00	5,980.00		13.50
DL	100.00	100.00	100.00		0.01

3 - 10/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,750.00	1,310.00	10,100.00		11.50
DL	100.00	100.00	100.00		0.01

3 - 05/14/2023: Corn, silage

3 - 05/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,490.00	750.00	5,210.00		11.00
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,830.00	760.00	5,770.00		12.50
DL	100.00	100.00	100.00		0.01

4 - 10/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,660.00	1,360.00	11,400.00		12.10
DL	100.00	100.00	100.00		0.01

4 - 05/15/2023: Corn, silage

4 - 05/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,360.00	770.00	4,730.00		12.40
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,330.00	950.00	6,530.00		12.10
DL	100.00	100.00	100.00		0.01

5 - 11/06/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage
Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 75.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,610.00	1,380.00	11,900.00		12.20
DL	100.00	100.00	100.00		0.01

5 - 05/11/2023: Corn, silage

5 - 05/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,920.00	880.00	5,580.00		14.10
DL	100.00	100.00	100.00		0.01

6 - 11/06/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,550.00	1,190.00	9,740.00		11.40
DL	100.00	100.00	100.00		0.01

6 - 05/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	780.00	5,200.00		10.70
DL	100.00	100.00	100.00		0.01

7 - 11/07/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,270.00	1,280.00	10,400.00		12.20
DL	100.00	100.00	100.00		0.01

7 - 05/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,670.00	920.00	6,190.00		9.68
DL	100.00	100.00	100.00		0.01

Field 13 - 02/02/2019: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,580.00	1,280.00	20,000.00		13.50
DL	100.00	100.00	100.00		0.01

Field 14 - 10/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,770.00	1,300.00	11,000.00		12.80
DL	100.00	100.00	100.00		0.01

Field 14 - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,420.00	910.00	5,440.00		10.10
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

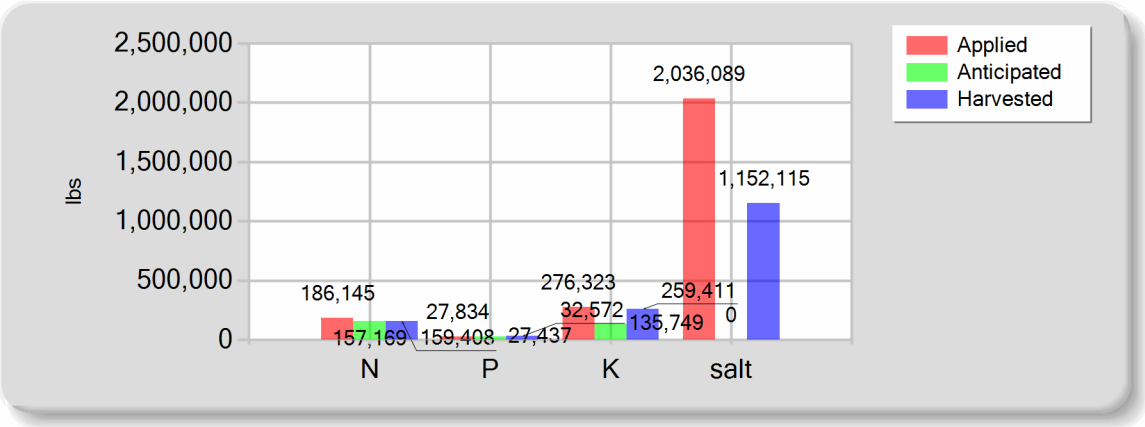
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

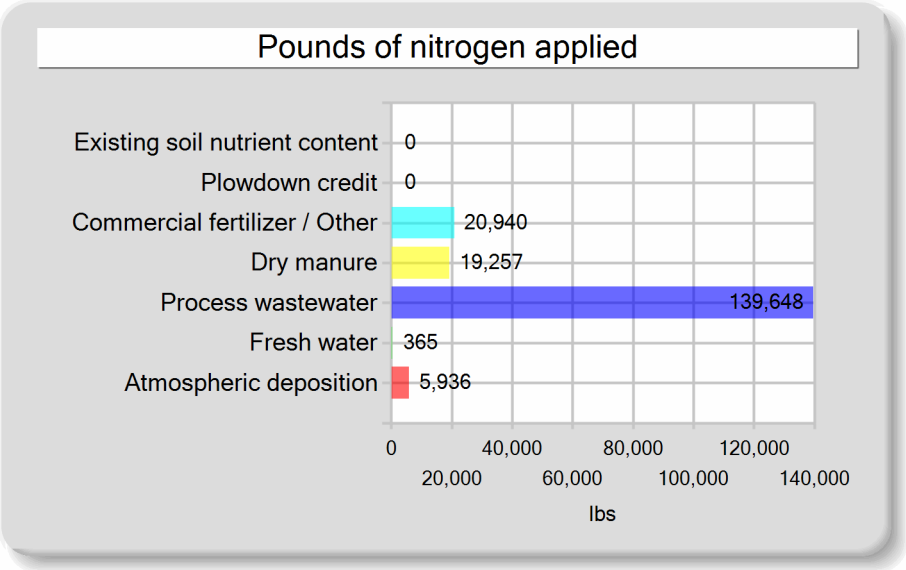
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

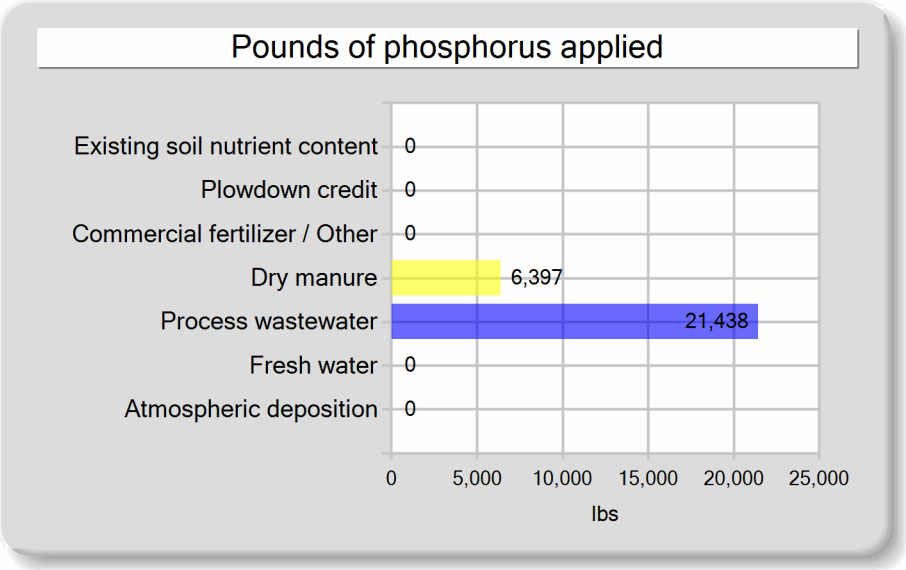
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	20,939.50	0.00	0.00	0.00
Dry manure	19,256.97	6,396.82	24,831.11	483,465.96
Process wastewater	139,648.15	21,437.65	251,492.32	1,437,372.01
Fresh water	364.78	0.00	0.00	115,251.07
Atmospheric deposition	5,936.00	0.00	0.00	0.00
Total nutrients applied	186,145.40	27,834.47	276,323.43	2,036,089.04
Anticipated crop nutrient removal	157,169.00	27,437.30	135,749.20	0.00
Actual crop nutrient removal	159,407.61	32,572.31	259,410.64	1,152,115.49
Nutrient balance	26,737.79	-4,737.84	16,912.79	883,973.55
Applied to removed ratio	1.17	0.85	1.07	1.77

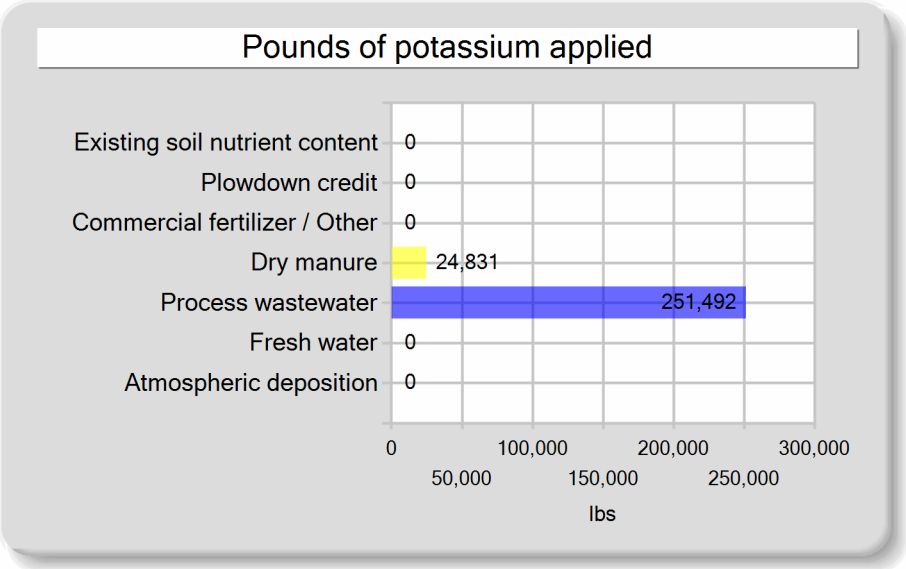
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

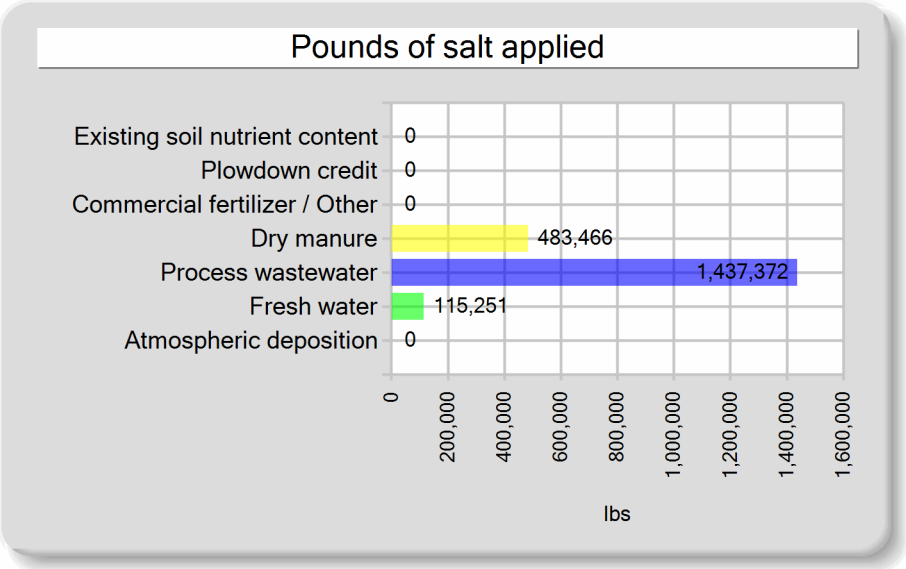


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field 6 Forage (1.45) had a slightly elevated field ratio as crop failed to meet anticipated removal of 170 pounds of nitrogen/acre. Crop yield was adequate however a low nitrogen value from the plant tissue lab analysis resulted in 9.1 pounds of nitrogen removed per harvested ton compared to the anticipated 10 pounds of nitrogen removed per harvested ton. Nitrogen application to the subject field of 224 pounds was not considered excessive but in line with that recommended in the dairy's nutrient management plan.

Field 3 (.93) and Field 4 (.84) both had field ratios below 1.00. The subject fields removed 258.9 and 291 pounds of nitrogen/acre compared to the anticipated 224 pounds of nitrogen/acre. Yields for the subject fields were below anticipated however high nitrogen lab values from the lab plant tissue analysis resulted in higher than anticipated removals. Nitrogen application to the subject fields was at levels recommended in the dairy's nutrient management plan.

Field 8 Forage, Field 9 Forage and Field 12 Forage were not harvested due to flooding from the San Joaquin River in January - April. Solid manure was applied to the fields prior to planting.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Edward Anthony Machado	Ed Jr. Machado
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Ed Machado

Name of Dairy Facility: Rivercrest Cattle

Facility Address: 8600 Dos Rios Lane, Modesto, CA 95358

Number and Street | City | Zip Code

Contact Person Name and Phone Number: Ed Machado Jr. 209-481-0482

Name | Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Rivercrest Cattle

Address of Hauling Company /Person: 11708 E. Moncure Road, Ripon, CA 95366

Number and Street | City | Zip Code

Contact Person: Ed Machado Jr. 209-481-0482

Name | Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

B&M Farms 11708 E. Moncure Road, Ripon, CA 95366 209-599-2565

Name | Number and Street | City | Zip Code | Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

1221 W. Lorenzen Road, Tracy, CA 95304

Number and Street | City | Zip Code | Assessor's Parcel Number

Dates Hauled: 4/15/2023 to 6/15/2023

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 1825 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 23.4% moisture

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

73 loads at 25 ton per load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 3/25/24

Hauler's Signature: _____ Date: 3/25/24

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Ed Machado

Name of Dairy Facility: Rivercrest Cattle

Facility Address: 8600 Dos Rios Lane, Modesto, CA 95358

Number and Street | City | Zip Code

Contact Person Name and Phone Number: Ed Machado 209-481-0482

Name | Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Rivercrest Cattle

Address of Hauling Company /Person: 11708 E. Moncure Road, Ripon, CA 95366

Number and Street | City | Zip Code

Contact Person: Ed Machado 209-481-0482

Name | Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

B&M Farms 11708 E. Moncure Road, Ripon, CA 95366 209-599-2565

Name | Number and Street | City | Zip Code | Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

1221 W. Lorenzen Road, Tracy, CA 95304

Number and Street | City | Zip Code | Assessor's Parcel Number

Dates Hauled: 5/15/23 to 10/30/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 6225 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 77.4% moisture

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

249 loads at 25 ton per load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 3/25 24

Hauler's Signature: _____ Date: 3/25/24

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Ed Machado

Name of Dairy Facility: Rivercrest Cattle

Facility Address: 8600 Dos Rios Lane, Modesto, CA 95358
Number and Street City Zip Code

Contact Person Name and Phone Number: Ed Machado 209-481-0482
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Rivercrest Cattle

Address of Hauling Company /Person: 11708 E. Moncure Road, Ripon, CA 95366
Number and Street City Zip Code

Contact Person: Ed Machado 209-481-0482
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

California Soils P.O. Box 345 Westley, CA 95387 209-835-9530
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

3401 Gaffery Road, Vernalis, CA 95385
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 7-1-23 to 11-2-23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 2949.9 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 9.4% moisture

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

126 loads at 23.41 ton per load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 3/25/24

Hauler's Signature: _____ Date: 3/25/24

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Rivercrest Cattle
Mailing address: 11708 E. Moncure Rd.
(not facility address) Ripon, CA 95366
Phone#:

Authorized Copy Released To: F+R Ag

Sampled By: Vince Furtado
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	11-29-23	8:30	DW #1 - Shiloh		15.0°C		✓				
2	11-29-23	8:45	DW #2 - Milk Barn		15.0°C		✓				
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	Vince Furtado	F+R Ag		11-29-23		
2nd		DENELE ANALYTICAL, INC.	11-29-23 12:45			T3333013

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Rivercrest Dairy
11708 E. Moncure Rd.
Ripon, CA 95366

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Cattle
RECEIVED DATE: 11/29/23 12:45 PM
SUBMITTED BY: Vince
ANALYZED DATE: 11/30/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW#1-Shiloh	T3333013A	Electrical Conductivity (EC)	EPA 120.1 and	1.10	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	3.45	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	704	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.8			
Electrical Conductivity (EC)	96.0			0.032
Nitrate Nitrogen (NO3-N)	93.8	102 (T3333002A)	98.3 (T3333002A)	3.56

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

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Rivercrest Dairy
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Grower

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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Cattle
RECEIVED DATE: 11/29/23 12:45 PM
SUBMITTED BY: Vince
ANALYZED DATE: 11/30/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW#2-Milk Barn	T3333013B	Electrical Conductivity (EC)	EPA 120.1 and	1.28	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	15.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	821	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.8			
Electrical Conductivity (EC)	96.0			0.032
Nitrate Nitrogen (NO3-N)	93.8	102 (T3333002A)	98.3 (T3333002A)	3.56

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